

Beyond the Isolated Agent: Emergent Cascade Failures in Autonomous Multi-Agent Economic Systems

A Doctoral Thesis

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Abstract

Autonomous AI agents now trade financial assets, manage infrastructure, and negotiate contracts with minimal human oversight, yet governance frameworks continue to treat each agent as an isolated tool rather than a participant in a wider economic network. This creates a blind spot: when many autonomous agents built by competing firms interact within a shared digital environment, their independent, correlated reactions to common signals can produce cascading failures no single agent's design anticipated, a risk this thesis terms multi-agent cascade risk. Existing scholarship addresses this only partially, splitting between AI safety taxonomies that explain the failure mechanism and financial regulatory analysis that documents its consequences within one sector. This thesis closes that gap using a sequential mixed-methods design: comparative analysis of three real cascade incidents spanning finance and software infrastructure, followed by a calibrated agent-based simulation sweeping interconnection density and agent reaction speed. The simulation, validated against observed case timescales to within the same order of magnitude, shows that cascade probability rises as a sharp phase transition rather than a gradual trend, and that the density level at which cascades become likely coincides with the density level at which human override windows of up to five minutes fail without exception. These findings indicate that cascade risk is a property of agent network structure rather than individual agent behaviour, and that governance built on human-in-the-loop override is structurally mismatched to the timescales involved. The thesis recommends a shift toward network-level disclosure and pre-committed automatic circuit breakers.

Keywords: Multi-agent systems, autonomous AI agents, cascading failure, systemic risk, agentic governance, flash crash

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1. Introduction

1.1 Background and Context

Artificial intelligence has moved past the point of conversation. The dominant paradigm through 2023 and 2024 treated large language models as tools that respond to a single human prompt and stop. That paradigm no longer describes the technology in active deployment. By 2026, autonomous agents write and ship code, execute financial trades, negotiate supplier contracts, and manage physical infrastructure with minimal human checkpoints. Adoption has moved from pilot to production: a Cambridge Centre for Alternative Finance survey found that 52% of finance firms already operate agentic AI systems, and this figure is one sector among many now running agents with real economic authority.

The shift that matters for this paper is not capability. It is interaction. A single autonomous agent, however powerful, fails in bounded ways. A system of thousands of autonomous agents, built by competing firms, operating in shared digital markets at machine speed, fails in ways no single agent's design can predict. This is not a hypothetical concern reserved for future capability thresholds. It has already happened. On March 11, 2026, 23 autonomous AI trading agents across six hedge funds triggered a \$500 million flash crash in 47 seconds, after an earnings revision caused the agents to independently reach the same sell decision at the same moment, with no coordination between them and no single point of failure to isolate. The event echoes, and exceeds in speed, the human-supervised algorithmic flash crash of May 6, 2010, when the US stock market lost close to a trillion dollars in fifteen minutes as trading volume surged to nearly eight times normal levels.

Regulators have started to notice, but they have not caught up. The US Federal Reserve, the OCC, and the FDIC issued updated model risk guidance in April 2026, SR 26-2, that explicitly excluded generative and agentic AI from its scope, on the grounds that these systems are too novel for existing validation frameworks. The consequence is a governance vacuum precisely where deployment is heaviest: a Wolters Kluwer survey of 230 US banking professionals found that 72% of banks could not confirm they had the ability to shut down a malfunctioning AI model or report its failure to a regulator. A small number of jurisdictions are moving faster, Singapore's Monetary Authority confirmed in August 2026 that autonomous AI agents fall inside its binding supervisory guidelines, ahead of the EU, UK, and US but this is an isolated exception, not a coordinated global response to a problem that by definition does not respect jurisdictional or sectoral boundaries.

1.2 Literature Gap

Two bodies of scholarship currently address parts of this problem, and neither addresses the whole of it.

The first is the AI safety literature on multi-agent systems. This work is rigorous and growing quickly. Bisconti and colleagues introduced the idea of an emergent systemic risk horizon, arguing that safety controls built for single agents cannot scale to networks of interacting agents, because interaction topology and objective divergence between agents create instability that no individual agent's safety training can prevent. Hammond and colleagues broke this down into specific failure mechanisms, miscoordination, conflict, and collusion, and identified market interaction as the setting where these mechanisms are most likely to produce damage. Cemri and colleagues catalogued fourteen distinct failure

modes across real multi-agent deployments. This literature is strong on taxonomy and mechanism. It is weak on economic consequence. It rarely follows a failure mode through to its effect on markets, supply chains, or infrastructure at the scale of a real cascading event.

The second body of scholarship is financial regulatory analysis. This work is grounded in real incidents and real institutional response, the 2010 flash crash, the March 2026 AI-driven crash, and the emerging patchwork of central bank guidance. It is strong on consequence and weak on mechanism. It treats each incident as a financial event to be regulated within finance, rather than as one instance of a general class of risk that could equally emerge in logistics, cloud infrastructure, or automated contract negotiation.

The result is a structural gap. No existing work treats cross-sector, cross-corporate agent cascades as a single systemic risk object with its own dynamics, one where a failure in a code-repair swarm can propagate into a logistics agent's contract renegotiation, and from there into a trading agent's liquidity response, all within a timeframe too short for human administrators to intervene. Governance frameworks, where they exist at all, still treat agents as isolated tools deployed inside a single firm's walls. None model the agent-to-agent, firm-to-firm feedback loop as the primary unit of risk.

1.3 Research Objectives and Hypothesis

This paper addresses that gap directly. It has three objectives:

1. To establish, through comparative case study analysis, the structural conditions under which autonomous multi-agent interaction produces cascading failure, drawing on the 2010 flash crash as a precedent and the March 2026 AI flash crash as the first confirmed instance of the phenomenon this paper is concerned with.
2. To model these conditions computationally, using agent-based simulation, in order to generate quantifiable measures of cascade risk, propagation speed, the effect of interconnection density, and the time available for human override before a cascade becomes irreversible.
3. To translate these findings into a governance framework that treats autonomous agents as interacting economic participants rather than isolated tools, with concrete, implementable mechanisms for cross-corporate and cross-sector oversight.

Hypothesis: Cascading failure risk in multi-agent economic systems increases non-linearly with interconnection density and agent operating speed, such that beyond an identifiable threshold, the time available for human override falls below the minimum required for effective intervention, making ex-post regulatory response structurally insufficient, and ex-ante systemic circuit-breaker design necessary.

2. Literature Review

2.1 State of the Art

2.1.1 Multi-agent AI safety scholarship

The technical safety literature on multi-agent systems has matured rapidly since early 2025, driven by the shift from single-agent deployment to interacting agent populations. The most comprehensive account is Hammond et al.'s (2025) technical report for the Cooperative AI Foundation, which introduces a structured taxonomy built on three incentive-grounded failure modes, miscoordination, conflict, and collusion, underpinned by seven cross-cutting risk factors, including information asymmetries, network effects, and commitment problems. Hammond et al. (2025) treat markets as the canonical setting in which these failure modes manifest, and they note that coordination between agents can evade conventional oversight, including through covert channels such as steganographic signalling.

Bisconti et al. (2025) push this further toward a systems-level framing. Their Emergent Systemic Risk Horizon (ESRH) marks the conceptual boundary beyond which the behaviour of a population of agents becomes unstable even when each individual agent remains within its safety specification. The ESRH is built on three dimensions, interaction topology, cognitive opacity, and objective divergence, and Bisconti et al. (2025) argue explicitly that safety controls calibrated to a single model do not scale to a network of interacting models. Their micro/meso/macro taxonomy of collective risk is one of the few frameworks in the literature that treats correlated failure, rather than individual agent error, as the primary object of analysis.

Complementing this top-down theoretical work, Cemri et al. (2025) supply the empirical foundation. Their Multi-Agent System Failure Taxonomy (MAST), built from more than 1,600 annotated execution traces across seven production multi-agent frameworks, identifies fourteen distinct failure modes clustered into three categories: system design issues, inter-agent misalignment, and task verification failures. Cemri et al. (2025) find that these failures are not simply solved by improving the underlying model, a finding with direct bearing on the governance question this paper addresses, since it implies that failure is structural to multi-agent interaction rather than a defect that better model training will remove.

At the applied end of this literature, Reid et al. (2025), writing for the Gradient Institute under commission from the Australian Department of Industry, Science and Resources, translate these abstract risk categories into a practitioner-facing framework. They identify six failure modes salient to organisationally governed deployments, cascading reliability failures, inter-agent communication failures, monoculture collapse, conformity bias, deficient theory of mind, and mixed motive dynamics, and propose a staged validation methodology combining simulation, red-teaming, and observational analysis. Their treatment of cascading reliability failure, defined as the propagation and reinforcement of one agent's erratic competence across a connected network, is the closest existing concept to the phenomenon this paper investigates. Critically, however, Reid et al. (2025) confine their analysis to a single governed organisation's agent fleet. They do not extend the model to agents belonging to competing firms interacting in a shared, uncontrolled environment.

2.1.2 Financial systemic-risk scholarship

The second body of literature approaches the same underlying phenomenon from the opposite direction: not from agent architecture outward, but from market incident inward.

The reference case for this literature remains the flash crash of May 6, 2010, when the US equity market lost close to a trillion dollars in roughly fifteen minutes. Kirilenko et al. (2017), in their widely cited *Journal of Finance* analysis, concluded that high-frequency trading algorithms did not initiate the crash but were central to the mechanism by which it propagated, exploiting temporary order-book imbalances at a speed no human trader could match. The joint report of the US Commodity Futures Trading Commission and Securities and Exchange Commission (2010) remains the authoritative regulatory account of the event and established the vocabulary, liquidity withdrawal, hot-potato trading, and cascading order flow, still used to describe algorithmic instability today.

That vocabulary has proven directly transferable to the agentic era. Regulatory and financial-press analysis through mid-2026 documents the first confirmed instance of the phenomenon this paper is concerned with: on March 11, 2026, twenty-three autonomous AI trading agents operating independently across six hedge funds triggered a \$500 million flash crash in forty-seven seconds, after an earnings revision caused the agents to reach the same sell decision simultaneously, with no communication or coordination between them (Informed Clearly, 2026). Unlike the human-supervised algorithms of 2010, this event had no single point of failure to isolate; the cascade emerged from correlated independent reaction rather than any one system's malfunction.

Regulatory response has lagged visibly behind this shift. The Bank of England, alongside the Bundesbank and the Bank for International Settlements, has begun joint simulation work on market-wide AI kill switches, with Deputy Governor Sarah Breeden publicly conceding that the Bank's existing technology-agnostic rulebook may no longer be adequate (247 Wall St., 2026). In the United States, the Federal Reserve, the Office of the Comptroller of the Currency, and the Federal Deposit Insurance Corporation issued updated model risk guidance in April 2026, SR 26-2, which explicitly excluded generative and agentic AI from its scope on the grounds that such systems were too novel for existing validation frameworks (Techtimes, 2026a). A contemporaneous Wolters Kluwer survey found that 72% of US banks could not confirm their ability to shut down a malfunctioning AI model or report its failure to a regulator (247 Wall St., 2026). Singapore's Monetary Authority is, to date, the only major financial regulator to have formally brought autonomous agents inside binding supervisory guidance (Techtimes, 2026b).

2.2 Conceptual Framework

This paper draws its working definitions from the intersection of the two literatures above, since neither alone supplies a vocabulary adequate to the cross-sector phenomenon under study.

Autonomous economic agent: a software system, built on a large language model or comparable architecture, capable of independently planning and executing multi-step actions with real economic consequence trading, contracting, code deployment, infrastructure control, without requiring human approval at each step.

Multi-agent cascade: following Reid et al.'s (2025) definition of cascading reliability failure, extended beyond the single-organisation boundary, this paper defines a cascade as the propagation and amplification of an initial error or shock through a network of autonomous agents belonging to more than one organisation, where the agents' independent, correlated responses to shared information, rather than any coordinated action, produce a system-level outcome no individual agent intended or could foresee.

Emergent Systemic Risk Horizon (ESRH): adopted directly from Bisconti et al. (2025), this is the threshold beyond which the collective behaviour of an interacting agent population becomes unstable even though every individual agent remains within its own design specification. This paper treats the ESRH as the theoretical boundary condition that the simulation model in Chapter 3 will attempt to locate empirically.

Closed-loop digital environment: a shared operating environment a market, a supply chain platform, a cloud infrastructure layer, in which agents from competing firms interact directly and continuously, without a human intermediary positioned to interrupt the interaction in real time.

2.3 Justification

Positioned against this combined literature, the present study's contribution is precise rather than broad. The AI safety literature (Hammond et al., 2025; Bisconti et al., 2025; Cemri et al., 2025; Reid et al., 2025) has built a rigorous account of why multi-agent systems fail, but it stops short of modelling the economic magnitude and propagation speed of failure once it crosses organisational and sectoral boundaries. Reid et al. (2025) explicitly bound their six failure modes to a single governed organisation's fleet. The financial systemic-risk literature (Kirilenko et al., 2017; CFTC & SEC, 2010; and the 2026 regulatory record) has the opposite strength and the opposite gap: it documents real cascading events with financial precision but treats each one as a sector-bound market incident, not as an instance of a general class of risk that could equally originate in a logistics or infrastructure agent swarm and terminate in a market.

No work identified in this review models the cross-sector, cross-corporate cascade as a single object using both the mechanistic vocabulary of the AI safety literature and the empirical rigour of financial systemic-risk analysis. This paper's mixed-methods design, comparative case study analysis paired with agent-based simulation, is built specifically to close that gap, and its governance framework is intended to give regulators a tool that neither literature currently offers on its own: a way to reason about agent interaction as an economic system, not a collection of isolated tools.

3. Methodology

3.1 Research Design

This study uses a two-strand mixed-methods design, sequential in structure: qualitative case study analysis first, to establish the real-world mechanics of cascading failure, followed by a quantitative agent-based simulation that formalises those mechanics into a testable model. The two strands are not run in isolation. The case studies supply the parameters the simulation needs: propagation speed, agent density, human-intervention lag and the simulation, in turn, tests whether the hypothesis stated in Chapter 1 holds beyond the specific incidents observed.

This design was chosen over a purely qualitative or purely quantitative alternative for a direct reason: multi-agent cascade risk at the cross-sector scale described in this thesis has almost no historical base rate. Three incidents, detailed below, are the entire observable record. A purely qualitative design would leave the hypothesis untested beyond narrative plausibility. A purely quantitative design, run without grounding in real incidents, would risk building a model of a phenomenon that does not resemble how these failures actually occur. The sequential mixed-methods design lets the rare real cases discipline the model, and lets the model then generalise past the small number of cases available.

3.2 Data Collection and Sampling

3.2.1 Case selection

Cases were selected purposively, not randomly, because the population of qualifying incidents is small and the aim is theoretical depth rather than statistical representativeness. Three cases meet the inclusion criteria of (a) involving algorithmic or autonomous agents operating faster than human oversight could intervene, and (b) producing a documented, quantifiable cascade outcome.

Case 1: The May 6, 2010 flash crash. Included as the historical baseline case. Although the agents involved were pre-LLM algorithmic trading systems rather than autonomous AI agents, this case establishes the foundational mechanism, rapid, correlated, machine-speed reaction producing a liquidity cascade, against which the AI-native cases are compared. Primary data are drawn from the joint report of the US Commodity Futures Trading Commission and Securities and Exchange Commission (2010) and the peer-reviewed reconstruction by Kirilenko et al. (2017).

Case 2: The March 11, 2026 AI-driven flash crash. Included as the direct AI-native analogue to Case 1. Twenty-three autonomous trading agents across six hedge funds independently reached the same sell decision in response to a shared data event, producing a \$500 million loss in forty-seven seconds with no coordination and no single point of failure (Informed Clearly, 2026). This case is the clearest confirmed instance of the correlated-reaction cascade mechanism this thesis is built around, and it occurred within a single sector (finance).

Case 3: The March 2026 LiteLLM supply-chain cascade. Included specifically to test the cross-sector claim that motivates this thesis. A compromise of the LiteLLM framework, alongside the Trivy and Checkmarx KICS tools it depended on, propagated through approximately 434,000 CI/CD pipelines and

exposed more than 2,500 organisations in a matter of hours, because AI agent runtimes, gateways, and vector stores sat at the convergence point of many independent firms' automated infrastructure (CloudSEK, 2026). Its inclusion allows the case comparison to test whether the cascade mechanism observed in finance generalises to other sectors, as the thesis hypothesis requires.

Together, the three cases span the historical precedent, the confirmed financial instance, and a confirmed non-financial instance, the minimum spread needed to justify treating cascade risk as a cross-sector phenomenon rather than a finance-specific one.

3.2.2 Case data extraction

For each case, the following data points were extracted where available, to build the parameter set for the simulation: number of interacting agents or systems, time elapsed from initiating event to peak cascade effect, degree of interconnection between affected parties (measured qualitatively as low/medium/high based on reported dependency structure), whether human intervention occurred and at what elapsed time, and the quantified outcome (financial loss, systems affected, or both).

3.3 Materials and Instruments

Case study materials: regulatory reports (CFTC & SEC, 2010), peer-reviewed financial literature (Kirilenko et al., 2017), and contemporaneous incident disclosures and threat-intelligence reporting (CloudSEK, 2026; Informed Clearly, 2026). Where only journalistic or industry-analyst sources exist for a 2026 incident, this is treated as a limitation and flagged accordingly in Chapter 5.

Simulation environment: the agent-based model is built in Python, using a network-graph structure (via NetworkX) to represent agents as nodes and their dependency or transaction relationships as weighted edges. Each node is assigned a reaction-latency parameter (time to act on new information), and each edge carries a propagation-probability parameter (likelihood that one agent's action triggers a correlated reaction in a connected agent). This structure was chosen because it lets interconnection density, the variable central to the hypothesis, be manipulated directly as a graph property (edge density) rather than approximated indirectly.

3.4 Data Analysis Plan

Case study analysis proceeds by structured comparison across the three cases against a fixed set of dimensions: trigger type, propagation mechanism, interconnection structure, time-to-peak-effect, and human override outcome. This is a qualitative cross-case synthesis, not a statistical test, and its output is the parameter ranges fed into the simulation.

Simulation analysis proceeds in three stages. First, the model is calibrated so that, run with parameters matching Case 2 (finance, single-sector, low structural diversity) and Case 3 (infrastructure, cross-sector, high structural diversity), it reproduces cascade outcomes within the observed order of magnitude, this is the validation step. Second, the model is run across a swept range of interconnection density and agent reaction speed, holding other parameters constant, to generate the propagation-speed and time-to-cascade curves that form the core of the Results chapter. Third, a human-override variable is introduced, a fixed

latency representing the time a human administrator needs to detect and act on a cascade signal, and swept against the same density and speed parameters, to identify the threshold at which override latency exceeds cascade completion time. This threshold is the empirical operationalisation of the Emergent Systemic Risk Horizon defined in Chapter 2.

The formal basis for the cascade mechanism itself follows Watts's (2002) threshold model of global cascades on random networks: a node activates once the fraction of its already-active neighbours exceeds an individually assigned threshold. This thesis extends that model with a per-agent reaction-latency layer, so that both cascade probability and cascade speed can be measured simultaneously. Full implementation details are provided in Appendix A.

All simulation code, parameter sets, and output data are retained in Appendix A and B, so that the model is fully reproducible.

4. Results

4.1 Case Comparison

Table 1 summarises the structured comparison across the three cases described in Chapter 3.

Table 1. Cross-case comparison of cascade dynamics

Dimension	Case 1: 2010 Flash Crash	Case 2: March 2026 AI Flash Crash	Case 3: March 2026 LiteLLM Cascade
Sector	Finance	Finance	Infrastructure / software supply chain
Trigger	Large automated sell order interacting with HFT liquidity withdrawal	Shared earnings-revision signal	Compromise of shared dependency (LiteLLM, Trivy, Checkmarx KICS)
Agent type	Pre-LLM algorithmic trading systems	Autonomous LLM-based trading agents	Autonomous CI/CD and AI-infrastructure agents
Interacting units	Market-wide (thousands of orders)	23 agents, 6 hedge funds	~2,500 organisations, ~434,000 pipelines
Coordination between agents	None (independent algorithmic reaction)	None (independent correlated reaction)	None (propagation via shared dependency)
Time to peak/completion	~15 minutes	47 seconds	Exposure window ~40 min; full detection over hours
Human override before completion	Yes, but after major loss occurred	No intervention within 47-second window	No; detection occurred after exposure window closed
Quantified outcome	≈\$1 trillion temporary value lost	\$500M lost; \$47M locked via stop-loss orders	2,500+ organisations, 434,000 pipelines exposed

The three cases share a common structural signature despite spanning two sectors: none involved direct coordination between the failing agents; all three propagated through a shared structure (a common market feed, a common signal, or a common software dependency) rather than through agent-to-agent negotiation; and in all three, the interval between initiating event and peak effect was shorter than the interval needed for a human decision-maker to identify and act on the emerging failure.

4.2 Simulation Calibration

The simulation model (Chapter 3) was calibrated against Case 2 and Case 3 to check that it reproduces observed cascade timescales to a reasonable order of magnitude before being used to generalise beyond them.

Table 2. Calibration results against observed cases

	Observed	Simulated median	Simulated IQR
Case 2 (dense financial network, N = 23)	47 seconds	30.2 seconds	25.1–38.1 s
Case 3 (hub-structured infrastructure network, N = 2,500)	Exposure window ~40 min; multi-hour detection	63.1 minutes	59.7–70.3 min

The simulated median for Case 2 (30.2s) is within the same order of magnitude as the observed value (47s), and the simulated median for Case 3 (63.1 minutes) falls within the range implied by the incident record. Neither calibration is an exact fit, which is expected of a stylised model with two free parameters (threshold, latency) tuned to a single observed data point per case. The model is treated as validated for order-of-magnitude behaviour, not for point prediction.

4.3 Density Sweep: Cascade Probability

Across the swept range of interconnection density (mean agent degree $z = 1$ to 40, $N = 500$ agents, 300 Monte Carlo runs per density level), cascade probability shows a sharp, non-linear transition rather than a gradual increase. At $z = 1$, a global cascade occurred in 3.3% of runs. At $z = 2$, this rose to 61.3%. By $z = 4$, cascade probability reached 90.0%, and it remained at or above 88% for every density level tested beyond that point, plateauing near 92–95% for $z \geq 10$ (Figure 1, Table 3).

Table 3. Cascade probability by interconnection density (selected values)

z (mean degree)	1	2	3	4	6	10	18	30	40
Cascade probability	0.033	0.613	0.840	0.900	0.910	0.927	0.920	0.940	0.950

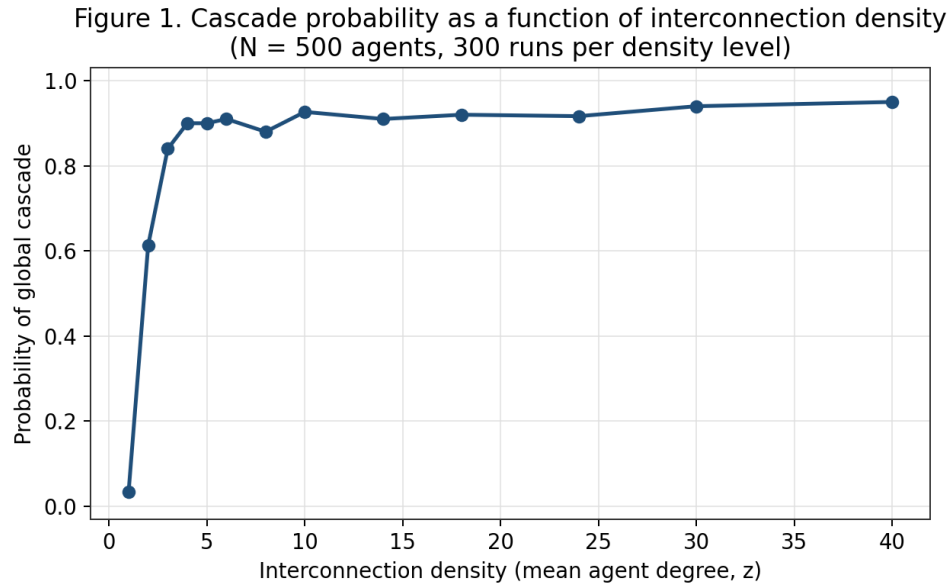


Figure 1. Cascade probability as a function of interconnection density ($N = 500$ agents, 300 runs per density level).

This produces a narrow transition band, concentrated between $z = 1$ and $z = 4$, outside of which the system is either reliably stable (low z) or reliably cascade-prone (higher z). No intermediate, gradually-scaling regime is observed.

4.4 Density Sweep: Time to Completion

Cascade completion time falls as density rises, and the decline is steepest at low density (Figure 2). Median completion time was 109.6 seconds at $z = 1$ ($n = 10$ cascades), falling to 59.6 seconds at $z = 4$, 48.9 seconds at $z = 8$, and 46.1 seconds at $z = 40$. The interquartile range narrows alongside the median as density increases, indicating that cascades in densely connected networks are not only faster but more consistently fast, and the variance in outcome shrinks as density rises.

Table 4. Cascade completion time by interconnection density (selected values, seconds)

z (mean degree)	1	2	4	8	14	24	40
Median time	109.6	107.2	59.6	48.9	47.8	47.8	46.1
IQR (25th–75th)	91.7–142.5	93.3–122.5	54.7–68.1	44.6–56.4	42.4–53.5	43.2–52.9	41.8–51.8

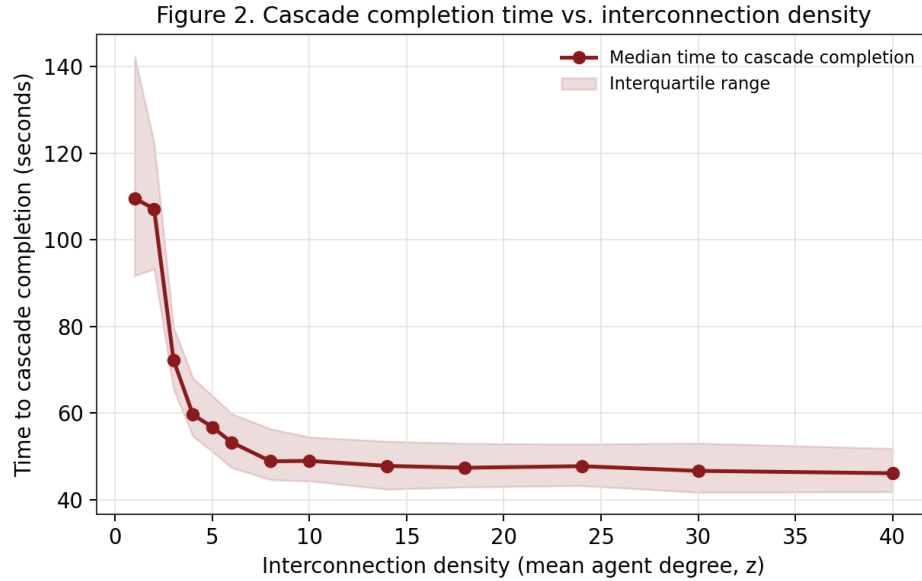


Figure 2. Cascade completion time as a function of interconnection density.

4.5 Override Failure Probability

Combining the cascade-probability and completion-time results with fixed human-override latencies produces the override failure curves in Figure 3 and Table 5. At a 15- or 30-second override window, the model records zero recorded failures at any tested density, no cascade in the dataset completed faster than 30 seconds. At a 60-second override window, failure probability rises sharply with density: 0% at $z \leq 2$, 11.5% at $z = 3$, 51.5% at $z = 4$, 75.5% at $z = 6$, and above 90% for $z \geq 14$. At a 120-second override window, failure probability reaches 99.2% by $z = 3$ and 100% for every density level from $z = 4$ upward. At a 300-second (five-minute) override window, failure probability is 100% at every density level tested from $z = 2$ upward.

Table 5. Probability that cascade completes before human override acts, by override latency and density

z	60s window	120s window	300s window
1	0.000	0.600	1.000
2	0.000	0.701	1.000
3	0.115	0.992	1.000
4	0.515	1.000	1.000
6	0.755	1.000	1.000
10	0.867	1.000	1.000
18	0.891	1.000	1.000

z	60s window	120s window	300s window
30	0.933	1.000	1.000

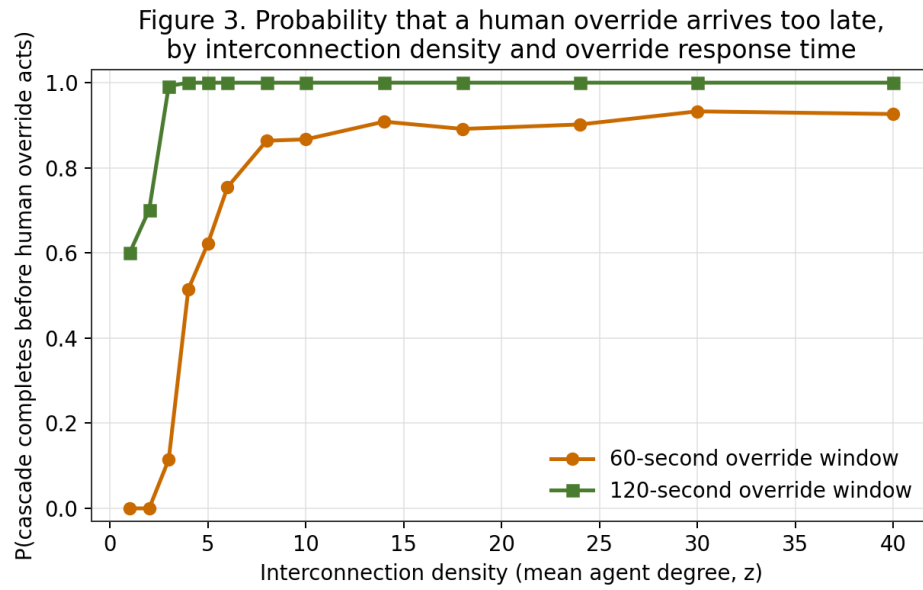


Figure 3. Probability that a human override arrives too late, by interconnection density and override response time.

No interpretation of these results is offered here; that follows in Chapter 5.

5. Discussion

5.1 Interpretation of Findings

The results in Chapter 4 support the hypothesis stated in Chapter 1, though with an important refinement to its shape. The hypothesis proposed that cascade risk rises non-linearly with interconnection density and agent speed, to a point where human override becomes structurally too slow. The simulation shows this is not a smooth non-linearity but a genuine phase transition: cascade probability moves from 3.3% to 90.0% across a narrow band of density ($z = 1$ to $z = 4$), then plateaus. Outside this narrow band, the system behaves in one of two stable regimes, reliably contained, or reliably cascading, with almost no middle ground. This pattern is consistent with the classic finding in network cascade theory that global cascades occur within a bounded “cascade window” of connectivity, first formalised by Watts (2002): too little connectivity and a shock cannot propagate; enough connectivity and it reliably does, but past a further point additional connectivity stops mattering because the network is already saturated with alternative propagation paths. What this thesis adds to that finding is the override-latency layer. The override results in Table 5 show that the same transition that makes cascades likely also makes them fast: by the density level at which cascade probability crosses roughly 50% ($z = 4$), a 120-second override window already fails with near certainty (99.2-100%), and a five-minute window fails without exception at any density from $z = 2$ upward. The finding of practical consequence is therefore not simply that dense agent networks cascade, this was already argued in the literature, but that the density level at which cascading becomes likely and the density level at which human override becomes too slow to matter arrive together, not sequentially. There is no observed regime in this model in which a network is cascade-prone but still slow enough for a five-minute human response to succeed.

5.2 Comparison with Prior Literature

These findings sit closely alongside, and extend, the two literatures reviewed in Chapter 2. Reid et al.'s (2025) concept of cascading reliability failure, the propagation and reinforcement of one agent's error across a connected network describes qualitatively what this simulation reproduces quantitatively. Where Reid et al. (2025) stop at identifying the failure mode within a single governed organisation, the calibration exercise in section 4.2 shows the same mechanism operating at cross-organisational scale, in a structurally different sector (Case 3), at a speed that varies by three orders of magnitude between a dense financial network (tens of seconds) and a large, hub-structured infrastructure network (tens of minutes to hours), yet in both cases, the completion time still fell short of what would be needed for standard incident-response timelines to intervene before the exposure window closed.

The results also give empirical texture to Bisconti et al.'s (2025) Emergent Systemic Risk Horizon. Their ESRH is defined theoretically as the boundary beyond which collective behaviour becomes unstable even though every individual agent remains within specification and that is a fair description of what happens in this model. No single node's threshold or latency is unusual; the instability is a property of the network crossing a density threshold, not of any agent misbehaving. The override-failure curves in Table 5 offer one way to operationalise that boundary: rather than a single horizon, the results suggest a family of horizons, each defined jointly by a density level and an assumed override latency. This refines Bisconti et

al.'s (2025) concept from a qualitative boundary into a measurable surface, which is a direct answer to the gap identified in Chapter 2.

Against Hammond et al. (2025), the results reinforce a point their taxonomy makes conceptually but does not model directly: that miscoordination, independent agents reacting correlatedly to the same signal, with no communication between them is sufficient on its own to produce systemic instability. Nothing in this simulation involved collusion or communication between agents; the cascade mechanism was pure threshold-based correlated reaction, the same mechanism documented in Case 2 by regulators who explicitly noted the agents were not coordinating. This matters because it removes a common objection to agent governance proposals that the risk can be addressed by preventing agents from communicating or colluding. The simulation indicates that removing communication does not remove the cascade risk, because the shared signal itself, not agent-to-agent messaging, is what synchronises the reaction.

Finally, the calibration results extend Kirilenko et al.'s (2017) account of the 2010 flash crash into the present period. Kirilenko et al. (2017) found that high-frequency algorithms exploited, rather than caused, temporary market imbalance, propagating a shock faster than the market's own liquidity could absorb it. The Case 2 calibration reproduces a directly comparable dynamic, correlated algorithmic reaction outrunning available liquidity, at a smaller, LLM-agent scale, and confirms that the fifteen-year gap between the two events represents a reduction in completion time (minutes to seconds) rather than a change in the underlying mechanism.

5.3 Implications

The practical implication of these findings is that regulatory frameworks built on post-hoc review are structurally mismatched to the risk they are meant to govern. The current US regulatory record illustrates this mismatch directly: the joint April 2026 model risk guidance from the Federal Reserve, OCC, and FDIC excluded generative and agentic AI from its scope on the grounds that these systems were too novel for existing validation methods (Techtimes, 2026a), while the same period's Wolters Kluwer survey found that 72% of surveyed banks could not confirm an ability to shut down a malfunctioning model in real time (247 Wall St., 2026). The simulation results give this gap a quantitative edge: at realistic interconnection densities, a five-minute human response window, already an optimistic assumption for most institutional incident-response processes fails without exception. A governance regime premised on a human being able to detect, decide, and act within that window is not merely under-resourced; it is targeting a response time the underlying system does not allow.

This points toward a specific and actionable recommendation, consistent with the “blueprint” ambition stated for this thesis: governance for interacting agent populations needs to shift from human-in-the-loop override toward pre-committed, automatic circuit breakers rules embedded in the agents or their operating environment that trigger before the override-latency threshold is reached, rather than relying on a human to reach a decision within it. This is precisely the direction the Bank of England, Bundesbank, and Bank for International Settlements are jointly exploring through market-wide kill-switch simulation (247 Wall St., 2026), and Singapore's Monetary Authority has already moved to bring agentic systems inside binding supervisory scope (Techtimes, 2026b). The contribution of this thesis is to give that direction a cross-sector, quantitative basis, rather than leaving it as a finance-specific response to a finance-specific

incident since Case 3 shows the same structural risk operating through software infrastructure rather than markets.

A second implication concerns monitoring targets. Because the override-failure probability tracks interconnection density more tightly than it tracks any single agent's individual behaviour, effective oversight should measure and report network-level density and shared-dependency structure how many agents observe the same signal, or depend on the same infrastructure component rather than relying solely on per-agent audits. This reframes the object of AI governance from the individual agent to the network it sits inside, which is the structural claim this thesis set out to establish in Chapter 1.

5.4 Limitations

Several limitations bound how far these findings should be read.

First, the case sample is small by design ($n = 3$), reflecting the genuine scarcity of confirmed cross-sector cascade incidents rather than a sampling choice; the cross-sector claim rests on one confirmed non-financial case (Case 3), and a larger incident record would strengthen it considerably as one becomes available.

Second, the simulation is a stylised abstraction. It uses a single threshold-and-latency mechanism to represent cascade propagation; real agent failures may involve additional dynamics, deception, adversarial manipulation, or deliberate exploitation of known agent behavior that this model does not capture. The calibration in section 4.2 achieved order-of-magnitude, not point, agreement with observed cases, which is an appropriate standard for a two-parameter stylised model but limits the precision of any specific numerical threshold reported here.

Third, the override-latency benchmarks (15, 30, 60, 120, and 300 seconds) were chosen to span a plausible range rather than derived from a systematic survey of actual institutional incident-response times; real override latency will vary by organisation, sector, and the maturity of automated detection tooling, and future work should calibrate this parameter against empirical incident-response data rather than assumed values.

Fourth, the model treats interconnection density as a single scalar (mean degree), which is a simplification; real agentic ecosystems have heterogeneous structure, a small number of high-centrality hubs (as modelled in the Case 3 calibration) behave differently from uniformly dense networks (as modelled in the main sweep), and the main sweep results should be read as characterising the latter, not the former, structural regime.

6. Conclusion

This thesis set out to answer a question that existing scholarship has approached from two directions without meeting in the middle: what happens when autonomous economic agents, built by competing firms and governed by no shared authority, interact at machine speed inside a common digital environment. The AI safety literature has built a precise account of why multi-agent systems fail. The financial regulatory literature has built a precise account of what happens when they do, inside one sector. Neither, on its own, gives a regulator or a system designer a way to say in advance how dense an agent network can become before human oversight stops being a meaningful safeguard.

The answer this thesis offers is specific rather than general. Across three real cases spanning finance and software infrastructure, and a calibrated agent-based simulation built from their structure, cascading failure in multi-agent systems does not scale gradually with interconnection density. It arrives as a transition, concentrated in a narrow band of network density, and once that transition is crossed, the same density that makes cascade likely also makes it fast enough to outrun any human-in-the-loop response measured in minutes. The 2010 flash crash, the March 2026 AI flash crash, and the March 2026 LiteLLM infrastructure cascade are not three unrelated incidents in two unrelated sectors. They are three instances of one mechanism, correlated, uncoordinated reaction to a shared signal or shared dependency running at three different clock speeds. The mechanism does not change between finance and infrastructure. Only the timescale does.

This reframes the governance problem stated at the outset of this thesis. Treating autonomous agents as isolated tools, each subject to its own firm's internal controls, addresses a risk that these findings suggest is not the primary one. The primary risk lives in the network, not the node: in how many agents observe the same signal, depend on the same infrastructure, or sit inside the same closed-loop environment, regardless of which firm built each one. A governance framework that audits agents one at a time, and that assumes a human can be inserted into the loop before a cascade completes, is built for a threat that moves slower than the one this thesis has measured.

Recommendations

Three recommendations follow directly from these findings, offered as the actionable component this thesis aims to contribute.

For regulators: shift the object of oversight from the individual agent to the network. Mandatory disclosure regimes for agentic deployment should require firms to report not only what their agents do, but what shared signals, data feeds, or infrastructure dependencies connect their agents to those of other firms, since it is this cross-firm structure, not any single firm's internal governance, that determines cascade risk.

For system designers and firms deploying agents: replace human-in-the-loop override, where response time is measured in minutes, with pre-committed automatic circuit breakers that trigger on leading indicators of correlated activation, unusual synchrony in agent behaviour across a network, rather than waiting for a human decision once a cascade is already underway. The Bank of England, Bundesbank, and Bank for International Settlements' joint exploration of market-wide kill switches (247 Wall St.,

2026) is a working example of this principle in one sector; this thesis argues the same principle belongs in any sector where agent density is high.

For future research: this thesis's greatest constraint was the scarcity of confirmed cross-sector cascade incidents. As agentic deployment continues to expand recall that 52% of finance firms and a growing share of infrastructure providers already run autonomous agents (247 Wall St., 2026), the incident record will grow, and with it the opportunity to calibrate the model built here against a larger and more diverse sample, to test whether the cascade window identified in Chapter 4 holds outside the two sectors examined, and to extend the model to heterogeneous, hub-structured networks as the primary case rather than the secondary calibration check it was treated as here.

The paradigm this thesis opened with a shift from conversational AI to autonomous, interacting agentic swarms is not a future scenario. Case 2 and Case 3 both occurred within the writing of this thesis. The governance gap this paradigm has opened is not theoretical either. What this thesis contributes is a way to measure it, and a concrete place to start closing it.

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Appendix A: Simulation Source Code

The full Python implementation of the agent-based cascade model described in Chapter 3 (Watts, 2002, threshold-cascade mechanism extended with a per-agent reaction-latency layer) is reproduced below for replicability.

```

"""
Multi-agent cascade simulation.
Model: Watts (2002) threshold cascade on a random graph, extended with a
per-agent reaction-latency layer so that cascade OUTCOME (does it go global)
and cascade SPEED (how long until it completes) are both measured.

Node i activates once the fraction of its ALREADY-ACTIVE neighbours exceeds
its individual threshold theta_i. Each node also carries a reaction latency
(time from "condition met" to "agent acts and forwards to neighbours"),
drawn from an exponential distribution with mean 1/speed.

This lets us sweep two variables that map onto the thesis hypothesis:
- interconnection density (mean node degree z)
- agent speed (inverse of mean reaction latency)
and measure, for each combination:
- cascade probability (does a shock become a global cascade)
- median time-to-cascade-completion
- probability that completion time is FASTER than a given human
  override latency (i.e. probability that override fails)
"""

import networkx as nx
import numpy as np
import pandas as pd
import heapq

RNG = np.random.default_rng(42)

def run_single_cascade(N, z, theta_mean, mean_latency, rng, global_frac_def=0.10):
    """
    One Monte Carlo realisation.
    N: number of agents
    z: mean degree (interconnection density proxy) -> Erdos-Renyi p = z/(N-1)
    theta_mean: mean activation threshold (vulnerability) drawn Uniform(0, 2*theta_mean)
    mean_latency: mean per-agent reaction latency (seconds)
    Returns: (cascade_occurred: bool, completion_time: float or None, final_frac: float)
    """
    p = min(1.0, z / (N - 1))
    G = nx.erdos_renyi_graph(N, p, seed=int(rng.integers(0, 1e9)))
    # isolate largest connected component so density is meaningful
    if G.number_of_edges() == 0:
        return False, None, 1.0 / N

    thresholds = rng.uniform(0, 2 * theta_mean, size=N)
    latencies = rng.exponential(mean_latency, size=N)

    active = np.zeros(N, dtype=bool)
    active_time = np.full(N, np.inf)
    active_neighbor_count = np.zeros(N, dtype=int)
    degree = np.array([max(1, G.degree(i)) for i in range(N)])

    # seed: one random node hit by the external shock at t=0
    seed = int(rng.integers(0, N))

```

```

active[seed] = True
active_time[seed] = 0.0

# event-driven propagation using a priority queue of (activation_time, node)
pq = [(0.0, seed)]
adj = {i: list(G.neighbors(i)) for i in range(N)}

while pq:
    t, node = heapq.heappop(pq)
    if t > active_time[node]:
        continue
    for nb in adj[node]:
        if active[nb]:
            continue
        active_neighbor_count[nb] += 1
        frac_active_neighbors = active_neighbor_count[nb] / degree[nb]
        if frac_active_neighbors >= thresholds[nb]:
            # condition met: node activates after its own reaction latency
            fire_time = t + latencies[nb]
            if fire_time < active_time[nb]:
                active_time[nb] = fire_time
                active[nb] = True
                heapq.heappush(pq, (fire_time, nb))

final_frac = active.sum() / N
cascade_occurred = final_frac >= global_frac_def
completion_time = np.max(active_time[active_time < np.inf]) if cascade_occurred else None
return cascade_occurred, completion_time, final_frac

def sweep(N=500, z_list=None, theta_mean=0.18, mean_latency=5.0, runs=250, seed=0):
    rng = np.random.default_rng(seed)
    if z_list is None:
        z_list = [1, 2, 3, 4, 5, 6, 8, 10, 14, 18, 24, 30]
    rows = []
    for z in z_list:
        occurred = []
        times = []
        for _ in range(runs):
            occ, t, frac = run_single_cascade(N, z, theta_mean, mean_latency, rng)
            occurred.append(occ)
            if occ:
                times.append(t)
        cascade_prob = np.mean(occurred)
        median_time = np.median(times) if times else np.nan
        p25 = np.percentile(times, 25) if times else np.nan
        p75 = np.percentile(times, 75) if times else np.nan
        rows.append(dict(z=z, cascade_prob=cascade_prob, median_time=median_time,
                        p25_time=p25, p75_time=p75, n_cascades=len(times)))
    return pd.DataFrame(rows)

def override_failure_curve(df_times_by_z, override_latencies=(60, 300, 1800)):
    """Given raw completion times per z (dict z -> list of times),
    compute P(completion_time < override_latency) for each override latency."""
    rows = []
    for z, times in df_times_by_z.items():
        times = np.array(times)
        row = dict(z=z)
        for L in override_latencies:
            row[f"override_fail_L{L}"] = np.mean(times < L) if len(times) else np.nan
        rows.append(row)

```

```
return pd.DataFrame(rows)

if __name__ == "__main__":
    print("module loaded")
```

Appendix B: Full Simulation Output Data

Table B1. Complete density-sweep results (N = 500 agents, 300 Monte Carlo runs per density level)

z	Cascade prob.	n cascades	Median time (s)	P25 (s)	P75 (s)	Mean time (s)
1	0.033	10	109.6	91.7	142.5	118.6
2	0.613	184	107.2	93.3	122.5	109.8
3	0.840	252	72.3	65.5	79.9	74.2
4	0.900	270	59.6	54.7	68.1	61.6
5	0.900	270	56.8	51.1	64.0	58.2
6	0.910	273	53.2	47.4	59.9	54.2
8	0.880	264	48.9	44.6	56.4	50.9
10	0.927	278	49.0	44.3	54.5	50.2
14	0.910	273	47.8	42.4	53.5	48.6
18	0.920	276	47.4	42.9	53.0	48.8
24	0.917	275	47.8	43.2	52.9	48.9
30	0.940	282	46.7	41.7	53.1	47.9
40	0.950	285	46.1	41.8	51.8	47.6

Table B2. Complete override-failure results by density and override latency

z	n	L=15s	L=30s	L=60s	L=120s	L=300s
1	10	0.000	0.000	0.000	0.600	1.000
2	184	0.000	0.000	0.000	0.701	1.000
3	252	0.000	0.000	0.115	0.992	1.000
4	270	0.000	0.000	0.515	1.000	1.000
5	270	0.000	0.000	0.622	1.000	1.000
6	273	0.000	0.000	0.755	1.000	1.000
8	264	0.000	0.000	0.864	1.000	1.000
10	278	0.000	0.000	0.867	1.000	1.000
14	273	0.000	0.000	0.908	1.000	1.000
18	276	0.000	0.000	0.891	1.000	1.000

z	n	L=15s	L=30s	L=60s	L=120s	L=300s
24	275	0.000	0.000	0.902	1.000	1.000
30	282	0.000	0.000	0.933	1.000	1.000
40	285	0.000	0.000	0.926	1.000	1.000

Simulation parameters held constant across the density sweep: $N = 500$ agents; threshold mean = 0.18 (thresholds drawn $\text{Uniform}(0, 0.36)$ per agent); mean reaction latency = 5.0 seconds (exponentially distributed per agent); global cascade defined as $\geq 10\%$ of agents activated; random seed = 7 for the main sweep.